

Technical Requirement Document

Here are the Technical Requirement Documents (TRDs) based on the provided Functional Requirement Documents (FRDs):

TR-DLT-001: Load Customer and Order Data from CSV to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table pipeline to read customer and order data from CSV files, perform data cleansing and transformation, and load it into Delta tables.

Target Cluster Configuration:

- Cluster Name: Databricks COE Cluster
- Databricks Runtime Version: 11.3.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: customerdata
- Location: `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`
- Format: CSV
- Description: Customer data
- Dataset Name: orderdata
- Location: `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`
- Format: CSV
- Description: Order data

Target Data Details:

- Output Name: customer
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer Delta table
- Output Name: order

- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order Delta table

Job Flow / Pipeline Stages:

- Read customer data from CSV
- Read order data from CSV
- Transform and load data into Delta tables

Data Transformations / Business Logic:

- Step: Add new column `TotalAmount` to `order` table
- Description: Calculate `TotalAmount` as product of `PricePerUnit` and `Qty`
- Transformation Logic: `TotalAmount = PricePerUnit Qty`
- Step: Remove null and duplicate records
- Description: Remove null and duplicate records from `customer` and `order` tables
- Transformation Logic: Use `dropna()` and `dropDuplicates()` functions

Error Handling and Logging:

- Log errors to a designated log file
- Implement retry mechanism for failed tasks

TR-DLT-002: Create and Load Order Summary Table

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a Delta Live Table pipeline to create an `ordersummary` table and load it with joined data from `customer` and `order` tables using SCD Type 2 logic.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer Delta table
- Dataset Name: order
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: ordersummary
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.ordersummary`
- Format: Delta
- Description: Order summary Delta table

Job Flow / Pipeline Stages:

- Create `ordersummary` table if it does not exist
- Join `customer` and `order` tables on `CustId` column
- Load joined data into `ordersummary` table using SCD Type 2 logic

Data Transformations / Business Logic:

- Step: Join `customer` and `order` tables
- Description: Join `customer` and `order` tables on `CustId` column
- Transformation Logic: Use `join()` function with `CustId` column
- Step: Implement SCD Type 2 logic
- Description: Maintain history in `ordersummary` table
- Transformation Logic: Use `merge()` function with `SCD Type 2` logic

Error Handling and Logging:

- Same as TR-DLT-001

TR-DLT-003: Update Order Summary Table on Customer Changes

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table pipeline to update the `ordersummary` table whenever there are changes to the `customer` table, using SCD Type 2 logic.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer Delta table

Target Data Details:

- Output Name: ordersummary
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.ordersummary`
- Format: Delta
- Description: Order summary Delta table

Job Flow / Pipeline Stages:

- Detect changes to the `customer` table
- Update `ordersummary` table by making old records inactive and new records active

Data Transformations / Business Logic:

- Step: Detect changes to `customer` table
- Description: Detect changes to `customer` table
- Transformation Logic: Use `change data feed` feature
- Step: Update `ordersummary` table
- Description: Update `ordersummary` table using SCD Type 2 logic
- Transformation Logic: Use `merge()` function with `SCD Type 2` logic

Error Handling and Logging:

- Same as TR-DLT-001

TR-DLT-004: Create and Load Customer Aggregate Spend Table**Related Functional Requirement(s): FR-DLT-004**

Objective: Implement a Delta Live Table pipeline to create a `customeraggregatespend` table and load it with aggregated data from the `ordersummary` table.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.ordersummary`
- Format: Delta
- Description: Order summary Delta table

Target Data Details:

- Output Name: customeraggregatespend
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customeraggregatespend`
- Format: Delta
- Description: Customer aggregate spend Delta table

Job Flow / Pipeline Stages:

- Create `customeraggregatespend` table if it does not exist
- Aggregate `TotalAmount` column from `ordersummary` table by `Name` and `Date` columns
- Load aggregated data into `customeraggregatespend` table

Data Transformations / Business Logic:

- Step: Aggregate `TotalAmount` column
- Description: Aggregate `TotalAmount` column by `Name` and `Date` columns
- Transformation Logic: Use `groupBy()` and `sum()` functions

Error Handling and Logging:

- Same as TR-DLT-001

TR-DLT-005: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-005

Objective: Implement all previous requirements using Delta Live Tables.

Target Cluster Configuration:

- Same as TR-DLT-001

Source Data Details:

- Same as previous TRDs

Target Data Details:

- Same as previous TRDs

Job Flow / Pipeline Stages:

- Implement all previous requirements using Delta Live Tables

Data Transformations / Business Logic:

- Same as previous TRDs

Error Handling and Logging:

- Same as TR-DLT-001